Lesson notes · for you

Puzzles with rules

Optimization · Level 1 · Lesson 3

Time	Materials
20 min	Paper and a pencil; A rubber

Learning goals

- Solve a 4 × 4 sudoku with its rules.
- Cross out what can't go in a square.
- Guess only when you're stuck, and undo wrong guesses.

Activity: Sudoku swap

1. Fill (6 min): On scrap paper, fill a 4 × 4 grid so every row, column and 2 × 2 box has 1, 2, 3 and 4. Check every row, column and box.
2. Hide (3 min): Draw an empty grid and copy only five or six of your numbers into it. Keep your full answer hidden.
3. Solve (8 min): Swap with a partner. Write small candidates in each square, cross out, and write in any square with one candidate left.
4. Share (3 min): Check the answers together. Did anyone need a guess? Which square was the hardest?

Use a pencil so you can rub out a wrong guess. No device needed.

Discussion

- Why did crossing out save so many guesses?
- What do you do at a dead end?
- Which fair rule would you add to your school timetable?

Check yourself

- One rule of sudoku + what I do when I'm stuck.

Try it this week

Make a 4 × 4 sudoku for a friend or someone at home: draw the grid, fill it in fully on scrap paper first, then copy only five or six of the numbers. Swap puzzles and solve them by crossing out. Did anyone need a guess?

Worksheet

Puzzles with rules

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Name _____ Class _____ Date _____

1 **You are solving a big sudoku. Every empty square has two candidates left, and no square has only one. What do you do?**

- ◯ Give up: it can't be solved
- ◯ Guess in one square, keep crossing out, undo if it goes wrong
- ◯ Write any number anywhere

2 **In a sudoku, a number can appear twice in the same column if it's in different boxes.**

- ◯ True
- ◯ False

3 **Put the steps for solving a sudoku in order.**

- ☐ Cross out the numbers in its row, column and box
- ☐ Undo a guess that leads to a dead end
- ☐ If you're stuck, guess where there are fewest candidates
- ☐ List the candidates of an empty square
- ☐ Write in a square that has only one candidate left

4 **Match each word to what it means.**

Candidate	• • • •	A try when nothing is certain
Rule	• • • •	A square where nothing fits
Guess	• • • •	A number that could still go in a square
Dead end	• • • •	No number twice in a row, column or box

5 **Complete the sentence.**

Rules _____ out what can't be, and you only _____ when you're stuck.

Word bank: guess · write · cross · give

✎ **Which puzzles or games do you know that have rules (a board game, a word game, a timetable)? How do the rules help you solve them?**

Check your answers

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1 **Guess in one square, keep crossing out, undo if it goes wrong**

When crossing out stops, guess in a square with the fewest candidates and keep crossing out.

2 **False**

Each rule must hold on its own: no number twice in a row, no number twice in a column, and no number twice in a box.

3 **Top to bottom: 2, 5, 4, 1, 3**

Rules first: list, cross out, write in what's certain.

4 **Candidate — A number that could still go in a square**
Rule — No number twice in a row, column or box
Guess — A try when nothing is certain
Dead end — A square where nothing fits

The rules cross out candidates; a guess is only for when you're stuck; and a dead end tells you to undo the guess.

5 **Rules cross out what can't be, and you only guess when you're stuck.**

Crossing out does most of the work; guessing is the last resort, and even then you can undo it.